AI 101 for Mechanical Engineers · Module 13

What a language model produces

Lesson 13.1

What you will learn

> Explain what a language model produces, distinguish LLMs, foundation and multimodal models, recognise diffusion models and GANs, and explain why generated outputs need independent verification.

Lesson overview

Module 13 · Lesson 1

What a language model produces

AI 101 for Mechanical Engineers

1 A language model continues a token sequence

`The` `pump` `'s` `head` `drops` `when` `flow` `rises` `…`

Blue: tokens supplied. Orange: a plausible continuation generated by the model.

Fluent output shows that text was produced, not that its equations, references or assumptions are correct.

2 Three names, three different questions

LLM	A large language model: many parameters, broad training.
Foundation model	Pretrained on broad data for reuse in downstream tasks. Broad pretraining does not prove accuracy for your application.
Multimodal model	Handles more than one information type, such as text and images. Handling images does not mean every detail is interpreted correctly.

3 Other generative families

Diffusion model



Learns to reverse a noising process: noise → structure.

GAN



A generator and a discriminator are trained against each other.

Knowing the model family does not validate the dimensions or physics in a generated image.

4 Check generated output before use

Explanation: can contain a wrong equation, an invented reference or a missing assumption.

Concept drawing: can fail dimensional, manufacturing or safety requirements.

Standards: generated text is no substitute for the applicable engineering standard.

Generated code is only text → Execute with a tool → Test numbers and assumptions

Key takeaway

Generated output is a draft, not evidence.

Check equations, sources, assumptions and requirements; execute and test generated code before trusting its result.

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A language model continues token sequences



- Learns token-sequence patterns; can generate or transform text [67]
- LLM: large language model, typically many parameters, broad training
- A continuation alone does not verify, execute code or consult standards

Keep the output and its evidence separate

Fluent output shows

- Text was produced coherently

Still needs checking

- Equations
- References
- Assumptions

Three names answer different questions

Term	Describes	Limit
LLM	Large language model	Task-specific evaluation still needed
Foundation model	Broad pretraining, reuse downstream	Not proven for every engineering application
Multimodal model	More than one information type	No guarantee every detail is correct

> Foundation model: pretrained on broad data for downstream reuse [68]. Capabilities can vary by task.

Other generative families

Diffusion model

- Learns a reverse process from noised examples [69]
- Noised data toward a structured output

GAN

- Trains a generator and a discriminator [70]

Naming the family does not validate depicted dimensions or physical properties.

Generated explanations can be wrong

- May contain an incorrect equation [71]
- May invent a reference or omit an assumption [71]
- Polished wording does not fix these problems
- Check the equation, the source and the assumptions

A concept drawing is not a verified design

- It may fail dimensional, manufacturing or safety requirements
- Usefulness as an illustration is a separate question
- Generated text cannot replace an applicable engineering standard
- Synthetic results do not validate real equipment

Generating, executing and checking code are separate



- A code block is not evidence that it ran
- Execution alone does not confirm assumptions or the answer

CHECK YOUR UNDERSTANDING

Does generating code mean that the code was executed successfully?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does generating code mean that the code was executed successfully?

> No. Generation produces source text. A tool must execute it, and independent tests must check the numerical result and assumptions.

CHECK YOUR UNDERSTANDING

Why does calling a model multimodal not establish that a generated concept drawing is suitable for manufacture?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why does calling a model multimodal not establish that a generated concept drawing is suitable for manufacture?

> Multimodal describes the information types handled. A drawing may still fail dimensional, manufacturing or safety requirements, which need independent engineering checks.

Key takeaways

- Language models learn token patterns and can generate or transform text
- Foundation models: broad pretraining; multimodal: several information types
- Diffusion models and GANs are other generative families
- Check facts, assumptions and requirements in explanations and drawings
- Generated code must be executed and independently tested

References

[67] T. B. Brown et al., “Language models are few-shot learners,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

[68] R. Bommasani et al., “On the opportunities and risks of foundation models,” 2021, arXiv:2108.07258. doi: 10.48550/arXiv.2108.07258

[69] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 6840–6851.

[70] I. J. Goodfellow et al., “Generative adversarial nets,” in Adv. Neural Inf. Process. Syst. (NIPS), vol. 27, 2014, pp. 2672–2680.

[71] Z. Ji et al., “Survey of hallucination in natural language generation,” ACM Comput. Surv., vol. 55, no. 12, pp. 1–38, 2023. doi: 10.1145/3571730

Numbers match the course reference list. The written lesson is the authoritative text.